Aaradhaya Sutrave

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EDUCATION

Texas A&M University (TAMU) | Expected Dec. 2026

- Bachelor of Science in Mechanical Engineering | College Station, TX
- Relevant coursework: Heat Transfer, Dynamics Systems and Controls, Thermodynamics, Fluid Mechanics, Materials and Manufacturing in Design, Computer-Aided Engineering

TECHNICAL SKILLS

Programming & Data Acquisition: MATLAB, Python, LabVIEW, Arduino Programming, Excel

Instrumentation & Electronics: Lock-in Amplifier, Oscilloscope, Function Generator, NSOM probe, circuit simulation

Design & Analysis: SolidWorks, ANSYS, Fusion 360, Orca Slicer, 3D printing

EXPERIENCE

Undergraduate Research Assistant – Sy Bor Wen, Department of Mechanical Engineering May 2026 - Present

- Developed a custom LabVIEW VI for buffered and continuous data acquisition from a lock-in amplifier
- Configured lock-in amplifier sensitivity and low-pass filter settings to extract weak optical signals
- Worked with a function generator, oscilloscope, and NSOM probe to generate and measure optical signals near a microheater
- Contributed to an ongoing research manuscript on Near Field Radiation

Undergraduate Research Assistant – MIDL Research Lab Feb. 2026 - Present

- Developing MATLAB code to generate Voronoi diagrams and half edge mesh data structures that replicate Finite element analysis mesh
- Designing precision acoustic test apparatus (Kundt tube) parts using SolidWorks CAD; creating PVC tube flanges, speaker reducers and measurement ports. Then converted designs to G-code for 3D printing
- Contributed to an ongoing research manuscript on Noise-absorption of weave structures.

ENGINEERING PROJECTS

CNC Machining and CAM Project – Fusion 360 Apr. 2026

- Created a CAD model, generated CAM toolpaths and tool selection in Fusion 360, and machined a plastic part on a CNC mill

Backseat Minder – Sensor, Circuit, & Alert-System Design Jan. 2026

- Created circuit diagram and simulation for a self-charging monitoring concept with occupant, temperature, and humidity detection
- Designed system functionality for alerts through a cellular module and supported CAD work for PCB integration considerations

Aluminum Hammer Fabrication: Multi Process Manufacturing Nov. 2025

- Fabricated an aluminum hammer using manual milling, lathe turning and precision sanding in 4 - 5 weeks

Aerodynamic Car Design & Wind Tunnel Testing – SolidWorks, Flow Simulation Nov. 2025

- Led a four person team to design and test a low drag car body using SolidWorks and Flow Simulations.
- Validated the design through wind-tunnel testing and achieved one of the lowest coefficients.

Stepped Shaft FEA/Discontinuity Effects Nov. 2025 - Present

- Conducted FEA of load bearing stepped shaft under different types of loading and constraints using ANSYS and numerical code simulation. Evaluated bending, von Mises, and shear stress to assess the effects of geometric discontinuities.

Racing Wheelchair Frame Design and FEA Validation Feb. 2025

- Design lightweight frame in SolidWorks supporting human scale load and performed FEA analysis to verify stiffness and strength constraints

Mars Rover Simulation – Python Modeling & Optimization Oct. 2024

- With a team, built a Python simulation that parameterized rover mass, material properties, and DC motor behavior across randomized Mars-like terrain.
- Evaluated and optimized rover design parameters for environmental constraints

Heat Sink Manufacturing Design – CNC Process Planning Mar. 2024

- Selected material, cutting tools, machining parameters, and toolpaths for a CNC-machined heat sink. Estimated cycle time production capacity, and machine requirements from the manufacturing plan